

Infosys Springboard Virtual Internship 6.0 Completion Report

Batch Number : 209

Start date : 13-08-2025

Name : Pathrapalli Akash

Internship Duration: 8 Weeks

1. Project Title:

AI-Based Real Estate Evaluation System

2. Project Objective

The objective of this project is to develop an **AI-Based Real Estate Valuation System** that accurately predicts property prices using **machine learning algorithms**.

The system aims to analyze various property features such as location, area, quality, and year of construction to provide **data-driven price estimates**.

This project's goal is to:

- Help **buyers and investors** make informed decisions by understanding fair property values.
- Assist **real estate developers and analysts** in identifying market trends and pricing patterns.
- Demonstrate the **practical application of data science and machine learning** in solving real-world problems.

By leveraging models like **Linear Regression, Random Forest, and XGBoost**, the project contributes toward building a **smart, transparent, and efficient valuation tool** that aligns with Infosys Springboard's mission of enhancing employability through hands-on AI and data analytics learning.

3. Project description in detail

The **AI-Based Real Estate Valuation System** is a machine learning–powered web application designed to predict the selling price of a house based on various property attributes. The project integrates **data preprocessing, model training, and web deployment** to create an intelligent, interactive, and user-friendly prediction platform.

1. Data Collection:

The dataset used in this project contains multiple features such as living area (**GrLivArea**), overall quality (**OverallQual**), year built, and sale price. These features were used as the basis for model training and analysis.

2. Data Preprocessing:

- Missing values were handled by filling numeric columns with the mean and categorical ones with the mode.
- Duplicate records were removed to ensure data consistency.
- Categorical variables were converted into numerical format using **one-hot encoding** for model compatibility.

3. Exploratory Data Analysis (EDA):

- Generated **correlation heatmaps, histograms, and scatter plots** to visualize relationships between features and sale price.
- Identified that attributes such as **OverallQual, GrLivArea, and YearBuilt** were the strongest predictors of property price.
- Analyzed **trends** such as average price per year and price distribution across quality grades.

4. Model Development:

- Implemented multiple regression models — **Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor** — to identify the best-performing algorithm.
- Used **RandomizedSearchCV** for hyperparameter tuning to improve model performance and minimize overfitting.
- Evaluated models based on **RMSE, MAE, and R^2** scores, with **XGBoost** emerging as the most accurate model.

5. Model Deployment:

- Integrated the best model into a **Streamlit-based web application** for real-time property price prediction.
- The app includes modules for **price prediction, comparative market analysis (CMA), investment insights, and trend visualization**.
- Added an **Investment Bot** to assist users with market-related queries and insights.

Technologies Used

- **Programming Language:** Python
- **Libraries:** pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost, streamlit
- **Tools:** VS Code, Jupyter Notebook, GitHub, Streamlit Cloud

Real-World Impact

This project has significant real-world relevance in the **real estate domain**, where accurate property valuation is crucial. It enables:

- **Buyers and investors make** data-driven purchasing decisions.
- **Sellers and developers set** fair and competitive prices.
- **Market analysts** to forecast price trends based on historical data.

Overall, this project showcases the power of **machine learning and data visualization** in solving real-world business challenges through intelligent automation and predictive analytics.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understanding the internship objectives, identifying project scope, and collecting the housing dataset.	Dataset collected, project structure created, and initial data exploration completed.
Week 2	Perform data cleaning and preprocessing.	Handled missing values, encoded categorical data, and removed duplicates.
Week 3	Conduct exploratory data analysis (EDA) and visualize relationships.	Created correlation heatmaps, scatter plots, and histograms to identify key influencing features.

Week 4	Implement baseline machine learning models.	Developed and tested Linear Regression and Decision Tree models.
Week 5	Optimize models and perform hyperparameter tuning.	Trained Random Forest and XGBoost models, conducted RandomizedSearchCV for parameter tuning.
Week 6	Evaluate models and select the best-performing one.	XGBoost model selected based on lowest RMSE and highest R^2 score.
Week 7	Develop Streamlit web application and integrate model.	Created interactive UI with modules for prediction, trend analysis, and comparative market analysis.
Week 8	Perform testing, finalize documentation, and present the project.	Completed final testing, report preparation, and project presentation submission.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Initiation of the project, understanding requirements, and planning workflow.	Week 1
Prototype/First Draft	Developed baseline models and initial Streamlit UI for property valuation.	Week 4
Mid-Term Review	Reviewed model performance, UI progress, and integration of financial tools.	Week 6
Final Submission	Completed full system with advanced models, AI integration, visualizations, and financial modules.	Week 8
Presentation	Demonstrated the complete system, highlighting features, insights, and deployment.	Week 8

5b. Project execution details

1: Project Planning and Setup

During the initial stage, the team defined the project scope, identified objectives, and outlined the workflow. The required libraries and tools were installed, and a virtual environment was set up to maintain dependency consistency. A dataset

containing housing-related attributes such as area, year built, and overall quality was imported for analysis.

Phase 2: Data Preprocessing

The dataset underwent comprehensive preprocessing to ensure data quality and consistency:

- **Handling Missing Data:** Filled numerical missing values with column means and categorical values with mode.
- **Removing Duplicates and Outliers:** Duplicate records were removed to prevent bias.
- **Encoding Categorical Features:** Applied one-hot encoding to convert categorical attributes into numeric form.
- **Feature Selection:** Identified highly correlated features using correlation heatmaps for model efficiency.

3: Exploratory Data Analysis (EDA)

EDA was performed using **pandas**, **matplotlib**, and **seaborn** to visualize data trends and relationships:

- Correlation heatmaps identified strong predictors such as **OverallQual**, **GrLivArea**, and **YearBuilt**.
- Scatter plots and histograms revealed how features influenced sale price.
- Bar and box plots were used to analyze price variations by quality and condition.

4: Model Development and Evaluation

Multiple machine learning algorithms were implemented:

- **Models Used:** Linear Regression, Decision Tree, Random Forest, and XGBoost.
- Each model was trained and evaluated on 80-20 train-test splits using performance metrics — RMSE, MAE, and R^2 .
- **Hyperparameter tuning** was carried out using **RandomizedSearchCV** to optimize model accuracy.
- The **XGBoost model** achieved the best performance and was selected for deployment.

5: Model Deployment (Web App Integration)

The best model was serialized using **Pickle** and integrated into a **Streamlit web application** (`app.py`).

The app provides:

- **Real-time property price prediction**
- **Comparative Market Analysis (CMA)**
- **Investment Insights and Forecasts**
- **Correlation heatmaps and trend visualization**

A conversational **Investment Bot** was also implemented to answer user queries about market insights and predictions.

6: Testing and Final Submission

The app was thoroughly tested for functionality, accuracy, and usability. Minor UI enhancements and model refinements were made before final deployment.

Finally, the report, screenshots, and presentation were prepared and submitted as part of the internship deliverables.

6. Snapshots / Screenshots



The screenshot displays the 'AI-Based Real Estate Valuation System' web application. The interface is dark-themed with a teal header. It features a title 'AI-Based Real Estate Valuation System' with a house icon and a 'Deploy' button. Below the title, a prompt 'Enter property details to predict the house price:' is followed by several input fields and sliders. The inputs include 'Living Area (sqft)' (1500), 'Overall Quality (1 = Very Poor, 10 = Excellent)' (5), 'Overall Condition (1 = Very Poor, 10 = Excellent)' (5), 'Year Built' (2000), 'Garage Capacity (Cars)' (2), 'Number of Full Bathrooms' (2), 'Number of Bedrooms' (3), 'Total Rooms Above Ground' (6), and 'Lot Area (sqft)' (8000). A 'Predict House Price' button is located at the bottom left.

Property Detail	Value
Living Area (sqft)	1500
Overall Quality (1 = Very Poor, 10 = Excellent)	5
Overall Condition (1 = Very Poor, 10 = Excellent)	5
Year Built	2000
Garage Capacity (Cars)	2
Number of Full Bathrooms	2
Number of Bedrooms	3
Total Rooms Above Ground	6
Lot Area (sqft)	8000

localhost:8502

Navigation

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- Comparative Market Analysis (CMA)
- About

1500

Overall Quality (1 = Very Poor, 10 = Excellent)

Number of Full Bathrooms

Overall Condition (1 = Very Poor, 10 = Excellent)

Number of Bedrooms

Year Built

2000

Total Rooms Above Ground

Lot Area (sqft)

8000

Predict House Price

Estimated House Price: \$123,265.45

Download Prediction (CSV)

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Prediction History

	GrLivArea	OverallQual	OverallCond	YearBuilt	GarageCars	FullBath	BedroomAbvGr	TotRmsAbvGrd	LotArea	PredictedPrice
0	1500	5	5	2000	2	2	3	6	8000	123265.4453
1	1500	5	5	2000	2	3	4	5	8000	122516.4297
2	1500	6	5	2000	2	3	4	5	8000	130115.5
3	1500	5	5	2000	2	2	3	6	8000	123265.4453

Download Full History

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Navigation

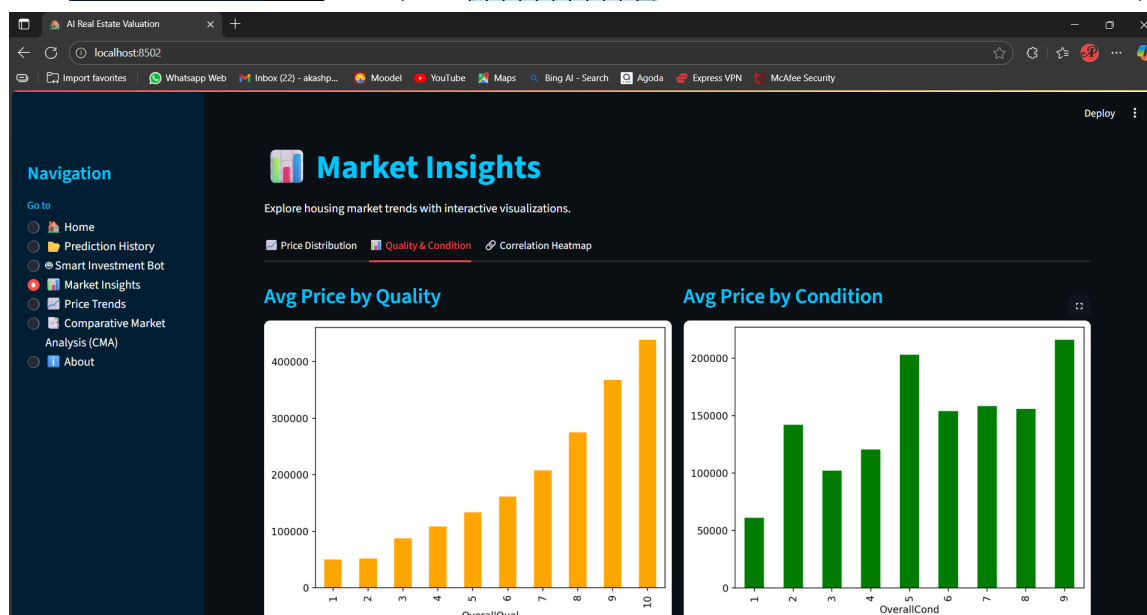
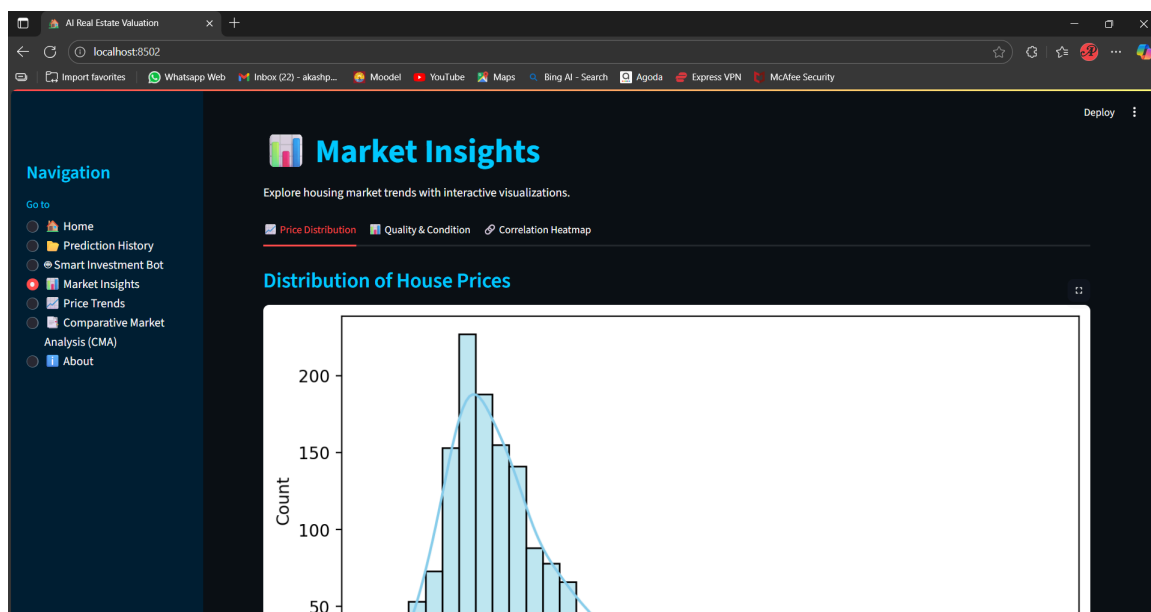
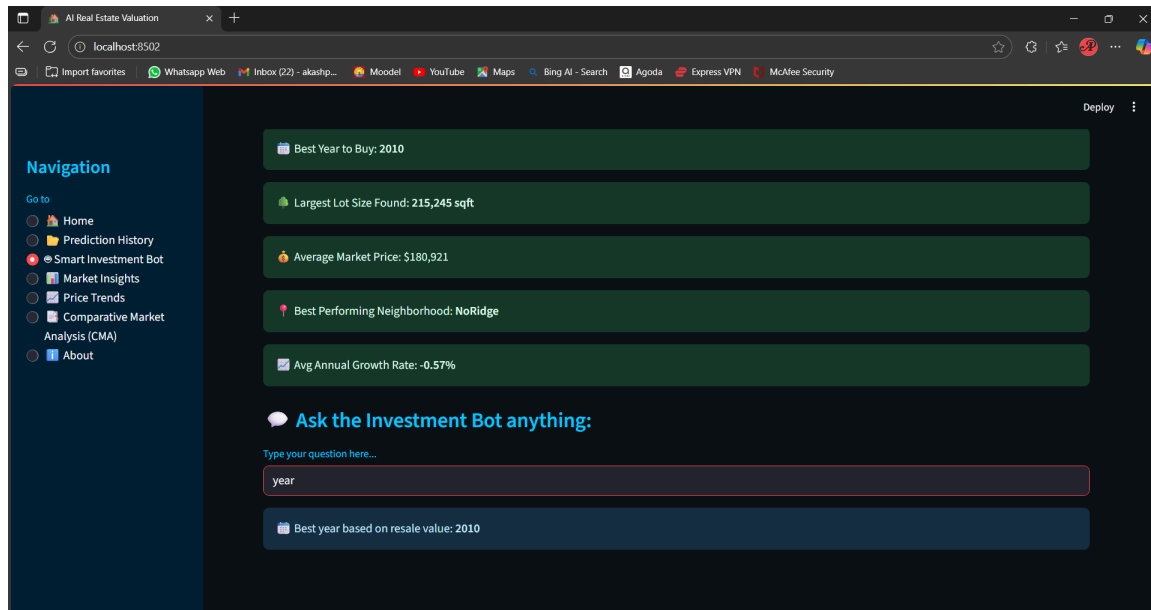
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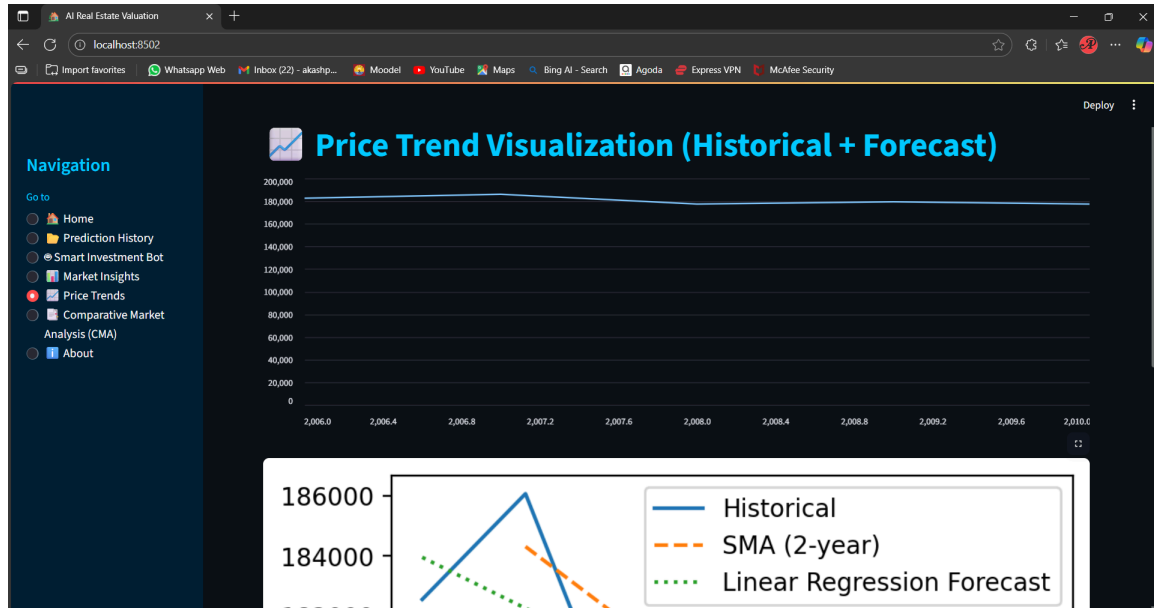
Smart Investment Bot

Get AI-powered insights for smarter real estate investment decisions.

Quick Insights:

- Best Quality Homes: Quality 10
- Best Year to Buy: 2010
- Largest Lot Size Found: 215,245 sqft
- Average Market Price: \$180,921
- Best Performing Neighborhood: NoRidge
- Avg Annual Growth Rate: -0.57%





AI Real Estate Valuation

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Comparative Market Analysis (CMA)

Compare multiple properties side by side and get AI-style investment suggestions.

Upload CSV with property details (columns must include GrLivArea, OverallQual, SalePrice, etc.)

Drag and drop file here
Limit 200MB per file • CSV

train.csv 449.9KB

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	LotConfig	LandSlope	Neighborhood	Condition1	Condition2	BldgType	House
0	1	60	RL	65	8450	Pave	None	Reg	Lvl	AllPub	Inside	Gtl	CollCr	Norm	Norm	1Fam	2Stor
1	2	20	RL	80	9600	Pave	None	Reg	Lvl	AllPub	FR2	Gtl	Veenker	Feedr	Norm	1Fam	1Stor
2	3	60	RL	68	11250	Pave	None	IR1	Lvl	AllPub	Inside	Gtl	CollCr	Norm	Norm	1Fam	2Stor
3	4	70	RL	60	9550	Pave	None	IR1	Lvl	AllPub	Corner	Gtl	Crawfor	Norm	Norm	1Fam	2Stor
4	5	60	RL	84	14260	Pave	None	IR1	Lvl	AllPub	FR2	Gtl	NoRidge	Norm	Norm	1Fam	2Stor
5	6	50	RL	85	14115	Pave	None	IR1	Lvl	AllPub	Inside	Gtl	Mitchel	Norm	Norm	1Fam	1.5Fin
6	7	20	RL	75	10084	Pave	None	Reg	Lvl	AllPub	Inside	Gtl	Somerst	Norm	Norm	1Fam	1Stor
7	8	60	RL	None	10382	Pave	None	IR1	Lvl	AllPub	Corner	Gtl	NWAmes	PosN	Norm	1Fam	2Stor
8	9	50	RM	51	6120	Pave	None	Reg	Lvl	AllPub	Inside	Gtl	OldTown	Artery	Norm	1Fam	1.5Fin
9	10	190	RL	50	7420	Pave	None	Reg	Lvl	AllPub	Corner	Gtl	BrkSide	Artery	Artery	2fmCon	1.5Un

Run CMA

AI Real Estate Valuation

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3	4	70	RL	60	9550	Pave	None	IR1	Lvl	AllPub	Corner	Gtl	Crawfor	Norm	Norm	1Fam	2Stor
4	5	60	RL	84	14260	Pave	None	IR1	Lvl	AllPub	FR2	Gtl	NoRidge	Norm	Norm	1Fam	2Stor
5	6	50	RL	85	14115	Pave	None	IR1	Lvl	AllPub	Inside	Gtl	Mitchel	Norm	Norm	1Fam	1.5Fin
6	7	20	RL	75	10084	Pave	None	Reg	Lvl	AllPub	Inside	Gtl	Somerst	Norm	Norm	1Fam	1Stor
7	8	60	RL	None	10382	Pave	None	IR1	Lvl	AllPub	Corner	Gtl	NWAmes	PosN	Norm	1Fam	2Stor
8	9	50	RM	51	6120	Pave	None	Reg	Lvl	AllPub	Inside	Gtl	OldTown	Artery	Norm	1Fam	1.5Fin
9	10	190	RL	50	7420	Pave	None	Reg	Lvl	AllPub	Corner	Gtl	BrkSide	Artery	Artery	2fmCon	1.5Un

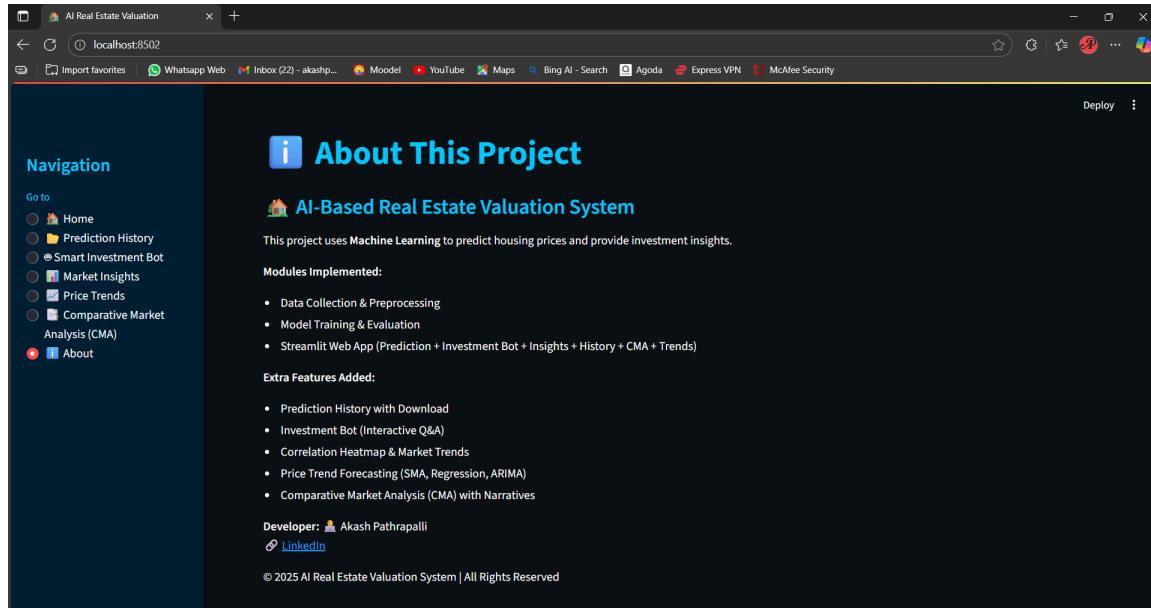
Run CMA

Best property has Quality 10 with price \$755,000

Investment Narrative

The average price of uploaded properties is \$180,921.
The property with the highest value has Overall Quality 10, built in 1994, and offers 4316 sqft living area.

✓ Suggestion: Prioritize properties with higher quality scores and larger living area, as they show stronger appreciation trends in future resale markets.



7. Challenges Faced

During the course of this 8-week internship project, several challenges were encountered at various stages of development. These challenges provided valuable learning experiences in data handling, model optimization, and deployment.

1. Data Quality and Missing Values

- **Challenge:** The dataset contained missing, inconsistent, and duplicate values which affected initial model accuracy.
- **Solution:** Implemented data cleaning techniques — filling missing numerical values with mean and categorical values with mode, and removing duplicates using pandas functions.

2. Feature Selection and Overfitting

- **Challenge:** The presence of many correlated features led to redundancy and model overfitting during early training stages.
- **Solution:** Used **correlation heatmaps** to identify and eliminate less significant features, and applied **cross-validation** and **hyperparameter tuning** to improve model generalization.

3. Model Optimization

- **Challenge:** Selecting the best algorithm and fine-tuning hyperparameters for optimal prediction accuracy required multiple iterations.
- **Solution:** Performed **RandomizedSearchCV** on Random Forest and XGBoost models to determine the best hyperparameters with minimal computational cost.

4. Tool Integration and Compatibility

- **Challenge:** During deployment, there were version mismatches between libraries (e.g., scikit-learn, xgboost, and streamlit) that caused runtime errors.
- **Solution:** Created a dedicated **virtual environment** and **requirements.txt** file to ensure smooth setup and consistent package management.

5. Streamlit App Deployment

- **Challenge:** Integrating multiple modules (prediction, CMA, trends, and investment bot) into a single Streamlit app initially caused performance issues.
- **Solution:** Optimized code structure, modularized functions, and used caching to improve app responsiveness and user experience.

8. Learnings & Skills Acquired

During the course of this 8-week virtual internship, I gained both **technical** and **professional** skills that enhanced my overall understanding of real-world data science and AI project development.

Technical Learnings

- **Data Preprocessing & Cleaning:** Learned techniques to handle missing data, detect duplicates, and encode categorical features for machine learning compatibility.
- **Exploratory Data Analysis (EDA):** Acquired skills to visualize and interpret data patterns using **matplotlib** and **seaborn**, identifying the most significant predictors of housing prices.
- **Machine Learning Model Development:**
 - Implemented and evaluated multiple regression models including **Linear Regression**, **Decision Tree**, **Random Forest**, and **XGBoost**.

- Understood how to use **performance metrics** like RMSE, MAE, and R^2 for model evaluation.
- **Hyperparameter Tuning:** Applied **RandomizedSearchCV** for optimizing model parameters to enhance accuracy and reduce overfitting.
- **Model Deployment:** Integrated the best-performing model into a **Streamlit web application**, learning about real-world deployment and interactive user interfaces.
- **Tools & Technologies Mastered:**
Python, pandas, numpy, scikit-learn, seaborn, matplotlib, xgboost, streamlit, GitHub

Analytical & Problem-Solving Skills

- Developed the ability to approach problems analytically, from understanding data to deriving actionable insights.
- Gained experience in transforming business problems into technical solutions through model-driven decision-making.

Soft Skills

- Improved **time management** and **collaborative skills** through team coordination and task distribution.
- Enhanced **communication skills** by explaining technical concepts clearly during discussions and reviews.
- Cultivated a mindset of **continuous learning**, **adaptability**, and **attention to detail** in project execution.

Overall Takeaway

This internship strengthened my foundation in **machine learning and data science**, improved my technical proficiency, and prepared me for **real-world AI-driven applications**. It also reinforced the importance of teamwork, documentation, and iterative improvement in project success.

9. Testimonials from team

My internship journey was both challenging and rewarding, providing hands-on exposure to real-world problem-solving and the practical application of machine learning.

The project allowed me to apply skills in **data preprocessing, feature engineering, model training, and deployment** within the context of real estate price prediction. I successfully executed an end-to-end AI project, integrating the **XGBoost/Random Forest backend** with a **Streamlit frontend** for real-time property valuation.

Key learnings included balancing technical depth with clear communication, managing milestones effectively, and refining **full-stack development skills**. The experience also strengthened soft skills such as **teamwork, time management, and structured problem-solving**, while offering comprehensive exposure to the complete machine learning workflow.

Overall, the internship was enriching, combining **technical rigor, practical implementation, and professional growth**, providing a strong foundation for future AI and software development endeavors.

10. Conclusion

The Infosys Springboard Virtual Internship 6.0 provided a valuable platform to gain **hands-on experience in machine learning, data analysis, and full-stack development**.

Through the development of the **AI-Based Real Estate Valuation System**, I successfully executed an end-to-end project — from **data preprocessing and exploratory analysis** to **model training, optimization, and web deployment**. The internship enhanced both **technical proficiency** and **professional skills**, including teamwork, time management, and effective communication.

This experience reinforced the application of theoretical knowledge in **real-world scenarios**, particularly in predicting housing prices and generating actionable market insights. Overall, the internship was highly enriching, equipping me with the **skills, confidence, and practical understanding** necessary for a career in AI, data science, and software development.

11. Acknowledgements

I would like to express my sincere gratitude to the **Infosys Springboard Internship Program** for providing me with this invaluable opportunity to work on the *AI-Based Real Estate Valuation System*. This internship allowed me to apply my theoretical knowledge of **Artificial Intelligence and Machine Learning** to a practical, end-to-end project. From the initial stages of **data preprocessing and exploratory data analysis** to **robust model development, comprehensive**

evaluation, and final deployment, I gained hands-on experience in every phase of the machine learning pipeline. This immersive experience significantly deepened my understanding of real-world AI applications and the challenges involved in bringing a concept to fruition.

I extend my heartfelt thanks to my mentor, **Ms. Thavayee Annadurai**, for her constant guidance, insightful feedback, and unwavering encouragement throughout the project. Her profound expertise and patient mentorship were instrumental in helping me navigate complex technical hurdles, refine my problem-solving strategies, and ultimately elevate the technical and analytical quality of my work. Her belief in my capabilities pushed me to strive for excellence and her advice was invaluable in shaping my approach to the project.

I also wish to thank my coordinator, **Mr. Praveen B**, for his valuable support and meticulous coordination. His efforts ensured a seamless and enriching internship experience, facilitating communication and providing the necessary resources for the project's successful execution. His organizational skills played a crucial role in maintaining a productive and structured learning environment.

Finally, I am deeply grateful to my peers for their collaborative spirit and willingness to share knowledge. The discussions and brainstorming sessions with them fostered a dynamic and engaging learning journey, contributing to a more comprehensive understanding of various approaches and solutions. This internship has not only significantly strengthened my technical foundation in AI and machine learning but also enhanced my problem-solving ability, critical thinking skills, and overall confidence to confidently undertake future AI-driven projects with a practical and informed perspective. The skills and insights gained here will undoubtedly serve as a strong foundation for my future endeavors in the field of Artificial Intelligence.